

Annotated Experiment Plan

Sampling Strategy and Statistical Defense for Clean-Label Malware Poisoning

RED NOTE - How to read this PDF

Black text is your proposed plan, reorganized for readability. Red text is my added warning, comment, or suggestion based on our previous discussion.

Overview

This experiment plan studies how sampling strategy affects both the effectiveness and the detectability of clean-label poisoning attacks against malware classifiers. The central idea is that recent work shows that not all benignware is equally useful for poisoning, yet many existing defenses were evaluated under random poison-sample selection rather than defense-aware sample selection.

The proposed defense is not a completely new defense family, but a targeted adaptation of existing statistical sanitization ideas to the clean-label malware setting. The specific contribution is to evaluate a retraining-free local statistical defense based on SHAP-space clustering with Gaussian Mixture Models (GMMs) and per-cluster Mahalanobis scoring, then analyze how its behavior changes under different poison sampling strategies.

RED NOTE - Good framing

This is the strongest part of the plan. Keep saying: this is not simply a new detector; it is a sampling-strategy-aware defense visibility study. This avoids overclaiming novelty against existing cluster-based mitigation work.

Research Questions

Primary question

How do poison-sample selection strategies affect the statistical detectability of clean-label malware poisoning under sanitization defenses?

Secondary questions

- Does malware-similar benignware selection make poisoned samples harder or easier to detect than random selection?
- Can a local subpopulation defense in SHAP space outperform global anomaly detection methods when the attacker uses structured sampling strategies?
- Does the proposed defense preserve rare or minority subpopulations better than broader pruning-based approaches?
- Which attack strategies, especially Combined, remain stealthy after local statistical filtering, and why?

RED NOTE - Narrow the questions

These questions are all useful, but for the final paper/thesis keep one main question and three subquestions. Treat rare-family preservation as an analysis item unless the dataset metadata makes it easy.

Motivation and Story

Severi et al. showed that malware classifiers can be poisoned by injecting benign-labeled files containing carefully chosen watermark features, causing malicious files with the same watermark to be misclassified as benign. Later work on PMSB showed that attack success depends not only on watermark design, but also on which benign samples are chosen for poisoning, because malware-similar benignware creates less conflict between benign structure and injected trigger features.

This creates an important research gap: the attack community now knows that sample selection matters, but the defense community has not systematically studied whether different sampling strategies leave different statistical footprints. A defense that works well against randomly selected poisoned benignware may fail when the attacker deliberately selects boundary-like or malware-similar benignware.

The proposed study connects three ideas: SHAP-guided clean-label poisoning from Severi-style attacks, benign subpopulation structure emphasized by PMSB, and local anomaly detection using cluster-aware distance measures. The resulting story is not simply "a better detector," but rather "a study of how attack sampling changes defense visibility, using a practical retraining-free statistical sanitizer."

RED NOTE - Excellent thesis story

This is defensible even if your proposed method is not the strongest defense. Your win condition can be: sampling changes the defense footprint, and different defenses fail for different reasons.

Experimental Design

Datasets

Primary dataset: EMBER, because it is the standard benchmark for static PE malware detection and is already used in Severi-style attack literature. A secondary dataset such as DREBIN, BODMAS, or a similar malware benchmark can be added to test whether findings generalize and to study behavior on imbalanced or heterogeneous families.

RED NOTE - Scope control

Use EMBER first. Add a secondary dataset only after the core story works. EMBER 2024/BODMAS can become messy because of family structure, temporal drift, and heterogeneity. Useful, but not step one.

Target classifiers

At minimum, include one fast tree-based classifier such as LightGBM, since it is common in malware detection and supports efficient SHAP computation. If time permits, add one neural-network classifier such as EmberNN to test whether defense behavior depends on model family.

RED NOTE - Recommendation

Start with LightGBM only. Adding EmberNN before your method is stable will multiply runtime and debugging. Model-family comparison is useful, but optional.

Attack generation

- Combined
- MinPopulation
- CountAbsSHAP

Evaluate each attack at multiple poisoning rates, such as 1%, 2%, and 5%, so the study can distinguish low-rate stealthy attacks from stronger but more visible poisoning campaigns.

RED NOTE - Attack subset first

For the first complete run, use only Combined and MinPopulation. Combined is the hard stealthy case; MinPopulation is likely easier for anomaly defenses. Add CountAbsSHAP later.

Sampling Strategies

For each attack strategy, vary the rule used to choose which benign samples will be poisoned:

- Random benign selection, as the historical baseline.
- PMSB-feature similarity, which selects malware-similar benignware in feature space.

- PMSB-distribution similarity, which selects benignware using distribution-aware similarity measures.
- PMSB-contribution similarity, which uses contribution or explanation similarity such as SHAP-based measurements.

This factor is the core independent variable of the study because it tests whether the attack sampling policy changes the defense observable signal.

RED NOTE - Avoid sampler overload

Four sampling strategies are enough. Do not add more just because they are available. The paper is stronger if the sampling set is small, interpretable, and tied to clear hypotheses.

Proposed Defense: SHAP-space GMM-Mahalanobis

Core idea

The defense models the benign class as a set of local statistical subpopulations rather than one single global benign distribution. It identifies benign-labeled points that are unusual within their own local subpopulation using Mahalanobis distance after GMM-based clustering.

Full defense pipeline

1. Train a baseline malware classifier on the poisoned training set.
2. Extract all benign-labeled training samples from that set.
3. Compute SHAP vectors for those benign-labeled samples using the target classifier.
4. Optionally reduce SHAP dimensionality using PCA while retaining a high proportion of explained variance, such as 95%, to improve numerical stability.
5. Fit Gaussian Mixture Models with varying numbers of components and select the best model using BIC.
6. Assign each benign sample to its most likely GMM component.
7. Compute local Mahalanobis distance from each sample to its assigned component mean and covariance.
8. Standardize the distance score within each cluster, for example by z-score or percentile rank.
9. Flag the highest-scoring samples as suspicious and remove them from the training data.
10. Retrain the malware classifier on the sanitized training set and evaluate attack success and clean performance.

RED NOTE - Important implementation details

Use covariance regularization. Start with diagonal covariance for GMM. Cap PCA dimensions, e.g., keep 95% variance OR max 50/100 components. BIC can choose tiny unstable clusters, so enforce a minimum cluster size or at least report cluster sizes.

RED NOTE - SHAP source warning

Computing SHAP from the poisoned model is realistic, but it may reflect poisoned decision logic. If time permits, compare against SHAP from a clean/trusted baseline model. If not, state clearly that the defense assumes only a suspicious model/training set is available.

Why this defense is distinct

This defense differs from prior cluster-based mitigation because that method scores suspicious clusters using model loss and may require iterative retraining during sanitization. By contrast, the proposed method uses a statistical score after SHAP extraction, reducing computational cost and isolating whether poisoned samples are statistically abnormal inside benign subpopulations.

RED NOTE - Novelty wording

Do not claim "new clustering defense" too strongly. Safer wording: local explanation-space anomaly scoring for sampling-strategy-aware analysis. This acknowledges overlap with model-agnostic mitigation while keeping your distinct scoring signal clear.

Baselines for Comparison

Defense	Main idea	Why include it
---------	-----------	----------------

No defense	Baseline attack strength	Required reference point.
Isolation Forest	Global anomaly detection on benign samples	Strong generic anomaly baseline and Severi-style reference.
Global Mahalanobis	One benign distribution; equivalent to K=1	Essential ablation to show whether local modeling matters.
Spectral Signatures	Outlier scoring from dominant singular directions	Standard backdoor detection baseline; include if implementation is stable.
Nested Training or simplified disagreement ranking	Ensemble disagreement and progressive filtering	Good connection to malware-specific poisoning defenses; optional if runtime is high.
Proposed GMM-Mahalanobis	Local SHAP-space subpopulation modeling	Main method.

RED NOTE - Keep baselines minimal

Minimum defensible set: No defense, Isolation Forest, Global Mahalanobis, Local GMM-Mahalanobis. Add Spectral and Nested only if time permits or if already implemented. Do not let comparison hell eat the thesis.

Metrics

Attack-side metrics

- Attack Success Rate (ASR) before defense: percentage of backdoored malware samples classified as benign.
- Clean accuracy before defense: standard benign/malware classification accuracy on untouched test data.
- ASR after defense: residual attack strength after sanitization.
- Defense Survival Rate (DSR): ratio of post-defense ASR to pre-defense ASR, measuring how much of the attack survives.

Defense-side metrics

- Poison recall: fraction of true poisoned training samples removed by the defense.
- False positive rate on clean benign samples: fraction of clean benign training points wrongly removed.
- Clean accuracy after defense: standard test-set accuracy after retraining on sanitized data.
- Rare-family preservation: accuracy or retention on minority or underrepresented families, where available.
- Defense cost: runtime, number of retraining cycles, and memory cost.

RED NOTE - Metric priority

The must-have metrics are ASR before/after, DSR, poison recall, clean benign FPR, and clean accuracy after defense. Rare-family preservation is excellent but should be "where metadata is available," not a blocker.

Main Factorial Study

Factor	Levels
Attack strategy	Combined, MinPopulation, CountAbsSHAP
Sampling strategy	Random, PMSB-feature, PMSB-distribution, PMSB-contribution
Defense	Isolation Forest, Spectral Signatures, Nested Training baseline, Proposed GMM-Mahalanobis
Poisoning rate	1%, 2%, 5%
Dataset	EMBER primary, optional secondary dataset

RED NOTE - Do not run the full factorial first

Core first run: EMBER + LightGBM + Combined/MinPopulation + Random/PMSB-contribution + No defense/IF/Global-Maha/Local-Maha + 1% poison rate. Expand only after the patterns are interpretable.

Ablation Studies

- Feature-space ablation: compare SHAP vectors against raw feature vectors.
- Clustering ablation: compare GMM with fixed-K k-means or density-based clustering.
- Distance ablation: compare Mahalanobis distance with Euclidean distance inside clusters.
- Threshold ablation: compare z-score thresholding, percentile-based removal, and fixed top-p% pruning.
- Poison-rate ablation: repeat the defense at multiple poison rates.

RED NOTE - Ablation priority

Highest priority ablations: Global vs Local Mahalanobis, SHAP vs raw features, and top-p% thresholds. K-means/HDBSCAN ablations are nice but optional. Use top-p% removal as the primary threshold because it makes methods comparable under the same removal budget.

Analysis of Failure Cases

Failure analysis is essential because the most important result may not be "the defense always wins," but rather which kinds of poisoning remain stealthy and why. In particular, the Combined strategy may survive better because it was designed to embed watermark values in regions already supported by real goodware, making poisoned benign samples less anomalous globally.

The study should analyze the distribution of Mahalanobis scores for clean and poisoned points across sampling strategies. If PMSB pushes poisoned samples closer to the malware boundary, it may make them easier to detect locally; if it instead creates coherent borderline benign clusters, it may make them harder to separate from rare clean benignware. Distinguishing between those outcomes is one of the main scientific payoffs of the study.

RED NOTE - This is your safety net

Even if Local GMM-Mahalanobis is not the strongest defense, the failure analysis can still be the paper contribution: it explains how sampling strategy changes the defense landscape and why Combined remains hard.

Expected Contributions

- Empirical contribution: systematic evaluation of how poison-sample selection strategies affect defense detectability in clean-label malware poisoning.
- Method contribution: practical retraining-free statistical sanitizer based on SHAP-space GMM clustering and local Mahalanobis scoring.
- Analytical contribution: explanation of which attack and sampling combinations survive local sanitization, and whether structured sample selection trades off attack strength against detectability.

Even if the proposed defense is not the strongest in absolute performance, the study is still valuable if it clearly explains when local statistical defenses succeed, when they fail, and how sampling strategy changes the defense landscape.

RED NOTE - Best contribution wording

Phrase the method contribution as "a lightweight statistical sanitizer and analysis tool," not as a final robust defense. The empirical and analytical contributions are likely your strongest claims.

Execution Order

11. Reproduce the baseline clean classifier on EMBER.
12. Reproduce Severi-style attacks under random sampling.
13. Reproduce or implement PMSB sampling strategies.
14. Confirm attack-side ASR differences across sampling strategies.
15. Implement the proposed GMM-Mahalanobis defense.
16. Run baseline defense comparisons on random sampling first.
17. Expand to the interaction study across sampling strategies.
18. Run ablations and rare-family preservation analysis.
19. Summarize qualitative patterns and failure cases.

RED NOTE - Add a hard stop point

After step 6, stop and inspect the pattern. If Local-Maha does not behave differently from Global-Maha or Isolation Forest, pivot to analysis of why local statistics fail instead of expanding to every dataset and poison rate.

Decision Criteria

- The proposed defense significantly reduces ASR under one or more PMSB sampling strategies while preserving clean accuracy.
- The study shows that some sampling strategies systematically weaken or strengthen specific defenses, revealing an attack-defense interaction that was previously unmeasured.
- The study explains why certain stealthy attacks, especially Combined, remain difficult to sanitize even with local subpopulation modeling.

RED NOTE - Realistic success definition
These criteria are healthy because they give you multiple ways to succeed. Your thesis does not need to prove "my defense beats all defenses." It only needs to show a clear, evidence-based attack-defense interaction.

Suggested Minimal Core Experiment

Item	Minimal setting
Dataset	EMBER only
Model	LightGBM
Attack strategies	Combined, MinPopulation
Sampling strategies	Random, PMSB-contribution
Defenses	No defense, Isolation Forest, Global Mahalanobis, Local GMM-Mahalanobis
Poison rate	1% first
Threshold	Top 1%, 3%, 5% removal budget

RED NOTE - My strongest practical advice
Finish this minimal core experiment before adding anything else. A complete small experiment with clean plots is better than a huge unfinished factorial monster. Research is not demon summoning; factorial designs should not require candles.

Final Comment

Overall, this plan is coherent and defensible. Its strongest framing is not that it invents a completely new defense, but that it studies how structured poison-sample selection changes statistical detectability under sanitization defenses. The proposed SHAP-space GMM-Mahalanobis method is a focused, lightweight way to test local benign-subpopulation anomaly behavior.

RED NOTE - Bottom line
Good plan. Main risk is scope creep, not idea quality. Keep the first experiment small, make the comparison minimal, and let the results decide whether to expand.